Chapter 13

Nature of Light

DEFINITIONS:

Optics	Branch of physics that studies the characteristics and behaviors of both visible and infrared light as well as vision.
Geometrical Optics	Geometric optics focuses on reflection and refraction through mirrors, glasses, and lenses, with the assumption that light travels in a straight line.
Physical Optics	Physical optics focuses on the wave aspect of light, studying interference, diffraction, and polarization.

NATURE OF LIGHT:

The nature of light was the highly debated topic in seventeenth century. Many theories were advanced to explain its nature. Few of them are

1. **Corpuscular Theory:** Light consist of Tiny particles called corpuscles which travel with the speed of light.
2. **Hygen's Wave Theory:** Light is travels in the form of waves.
3. **Maxwell Electromagnetic Theory:** Light consist of Electromagnetic waves.
   - Light is electromagnetic radiation with dual characteristics as both particles (photons) and waves, known as wave-particle duality.
   - It travels at a constant speed of about 300,000 kilometers per second (3×10^8 m/s) in a vacuum and interacts with matter, causing phenomena like reflection, refraction, and absorption.
   - Understanding light has transformed communication, imaging, and scientific research.
4. **Plank's Quantum Theory:** Light consist of packets of energy called quanta i.e particle.

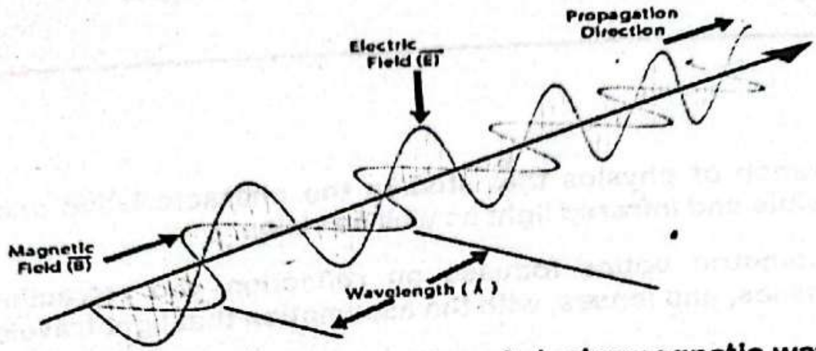
ELECTROMAGNETIC WAVES AND SPECTRUM:

- Transverse waves with electric and magnetic properties are known as electromagnetic waves.
- A changing magnetic field induces an electric field, according to Faraday's equations of electromagnetic induction, and James Clerk Maxwell established that a changing electric field also generates a magnetic field.
- Electric field lines in these waves run parallel to the plane of the page, while magnetic field lines run perpendicular to the plane.
- The interaction of these time-varying electric and magnetic fields produces electromagnetic waves that travel at the speed of light, about 3.00×10^8 m/s (c), as proved by Maxwell's use of Gauss' and Ampere's laws.

$$c = \frac{1}{\sqrt{\varepsilon_0\mu_0}}$$

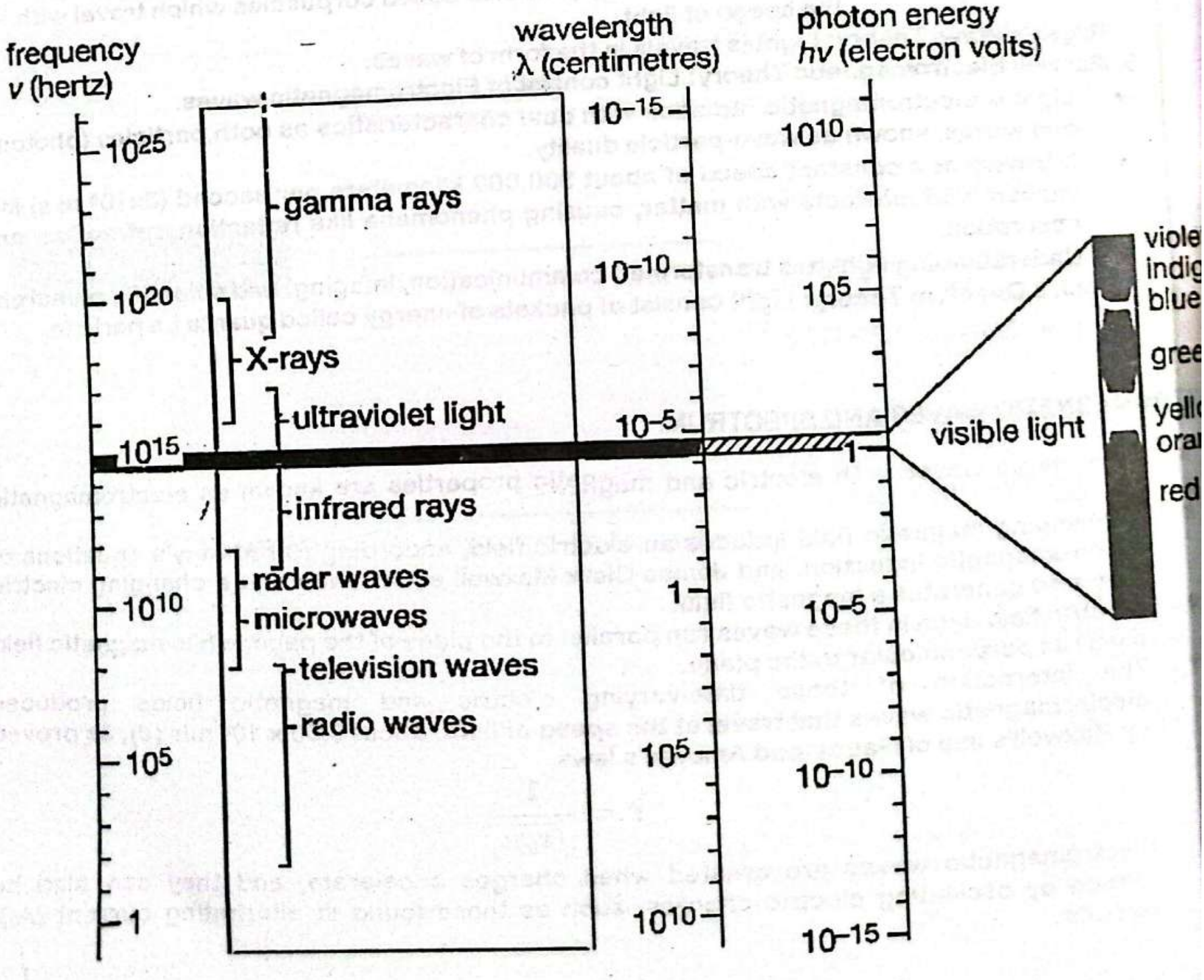
- Electromagnetic waves are created when charges accelerate, and they can also be formed by oscillating electric charges, such as those found in alternating current (AC) circuits.

Electromagnetic Wave



A simple depiction of travelling of electromagnetic waves

- **The electromagnetic spectrum is the wide range of electromagnetic wave frequencies and wavelengths.**
- **From radio waves to gamma rays, the electromagnetic spectrum is divided into seven bands. The regions borders are a little unclear and arbitrary.**



EM Spectrum	Discovery/Characteristics	Wavelength Range	Frequency Range
Radio Waves	Discovered by Heinrich Hertz in 1887.	1×10^{1} m to 3×10^{4} m	1×10^{7} Hz to 3×10^{10} Hz
Microwaves	Cosmic microwave discovery due to telescope noise.	3×10^{-1} m to 3×10^{-2} m	1×10^{10} Hz to 3×10^{11} Hz
Infrared	Discovered by Frederick William Herschel in 1880.	1×10^{-3} m to 1×10^{-1} m	3×10^{11} Hz to 3×10^{13} Hz
Visible Light	The range of light visible to the human eye.	3.8×10^{-7} m to 7.5×10^{-7} m	4×10^{14} Hz to 7.5×10^{14} Hz
Ultraviolet	Higher frequencies than visible light.	3.8×10^{-7} m to 1×10^{-8} m	7.5×10^{14} Hz to 3×10^{16} Hz
X-Rays	Discovered by Wilhelm Conrad Rontgen in 1895.	1×10^{-11} m to 6×10^{-11} m	5×10^{16} Hz to 3×10^{18} Hz
Gamma Rays	Emitted during atomic nucleus transitions.	Very short wavelengths	Very high frequencies

FOR VISIBLE SPECTRUM:

Color	Wavelength Range (m)	Frequency Range (Hz)
Red	$750\times10^{-9} - 610\times10^{-9}$	$4.0\times10^{14} - 4.9\times10^{14}$
Orange	$610\times10^{-9} - 595\times10^{-9}$	$4.9\times10^{14} - 5.0\times10^{14}$
Yellow	$595\times10^{-9} - 580\times10^{-9}$	$5.0\times10^{14} - 5.2\times10^{14}$
Green	$580\times10^{-9} - 500\times10^{-9}$	$5.2\times10^{14} - 6.0\times10^{14}$
Blue	$500\times10^{-9} - 480\times10^{-9}$	$6.0\times10^{14} - 6.3\times10^{14}$
Indigo	$480\times10^{-9} - 435\times10^{-9}$	$6.9\times10^{14} - 7.5\times10^{14}$
Violet	$435\times10^{-9} - 400\times10^{-9}$	$7.5\times10^{14} - 7.9\times10^{14}$

Chapter 13 — Wave Front

WAVE FRONT:

Whenever waves pass through a medium, its particles execute S.H.M (Simple Harmonic motion).The path (locus) of all the particles of the medium having the same phase in known as wave front.

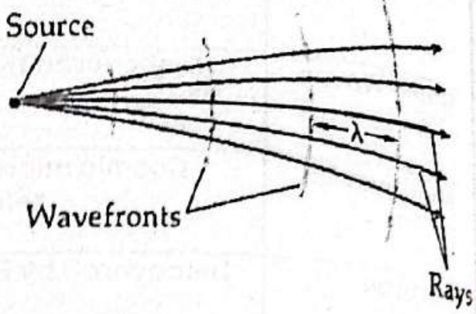
Characteristics:

1. It is at right angle to the direction of travel of waves
2. A line normal to the wave front indicating the direction of motion of waves called a ray.

TYPES OF WAVE FRONT

Spherical Wave Front

If a source of disturbance emit wave length in all directions then wave front of such waves are in form of concentric spheres such wave fronts are called spherical wave fronts.

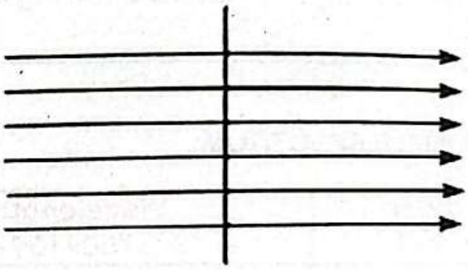


Circular Wave Front

When spherical wave fronts travel in a plane they are called circular wave fronts for example when stone drops in water the wave produced. It is two dimensional.

Plane Wave Front

Far from a source a small section of spherical wave front is nearly plane and is called plane wave front.



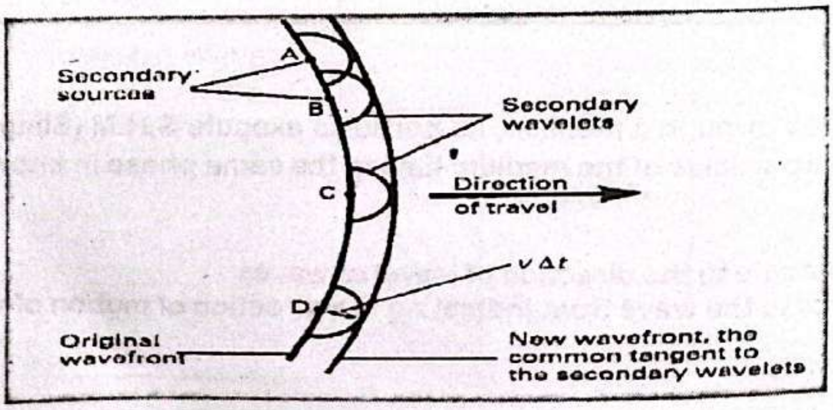
Plane wavefront

HUYGEN'S PRINCIPLE:

Huygens's principle is a geometrical method for finding the position of a new wave front from the known shape of a wave front at some instant.

Definition:

- Every point on a wave front can be considered as a source of secondary spherical wave front.
- The new position of the wave front after a time t can be found by drawing a plane tangential to the secondary wave-lets.



Application

- It is used to work out problem in diffraction and interference.
- It is very useful in the study of light.

INTERFERENCE OF LIGHT:

Definition

The phenomenon of two or more waves of the same frequency combining to form a wave in which the disturbance at any point is the algebraic or vector sum of the disturbances due to the interfering waves at those points".

There are two types of interference, named as

- Constructive Interference
- Destructive Interference.

Constructive Interference:

"If the crests of one wave fall on the crests of the other then these waves are said to interfere constructively".

OR

"If the resultant intensity of the interfering waves is greater than the intensity of an individual wave, then this type of interference is known as constructive interference".

Destructive Interference:

"If the crests of one wave coincide with the troughs of the second wave and vice versa and their displacement cancel at every point, then the two waves are said to interfere destructively."

OR

"If the resultant intensity of the interfering waves is zero or less than the intensity of the individual wave, then this type of interference is called destructive interference".

The conditions for interference are:

- The sources must be has coherent.
- The sources must be monochromatic.
- The superposition principle must apply.
- The two sources of wave should be very narrow and close

Mathamatically,

Condition for constructive interference

Constructive interference takes place when path difference between two waves is $0, \lambda, 2\lambda, 3\lambda$—— i.e.

$$\text{Path difference} = m\lambda \qquad \text{where } m = 0, 1, 2, 3 \ldots$$

In case of constructive interference the resultant intensity is increased and bright fringe is produced.

Condition for destructive interference

Constructive interference takes place when path difference between two waves is $\frac{\lambda}{2}, \frac{3}{2}\lambda, \frac{5}{2}\lambda$,——

i.e.

$$\text{Path difference} = \left(m + \frac{1}{2}\right)\lambda \qquad \text{where } m = 0, 1, 2, 3 \ldots$$

In case of destructive interference the resultant intensity is decreased and dark fringe is produced.

COHERENT SOURCES:

Definition

Those sources which are emitting light waves of same wave length, time period, frequency and amplitude. The phase difference should be either zero or constant

Method of producing coherent sources

A common method for producing two coherent light sources is to use one monochromatic source to illuminate a screen with two small slits

The light emerging from both slits is coherent because a single source produces the original light beam and the two slits serve only to separate the original beam into the parts. Consequently, a random change, in the light emitted by the source will in the two separate beams at the same time, and interference effects can be observed.

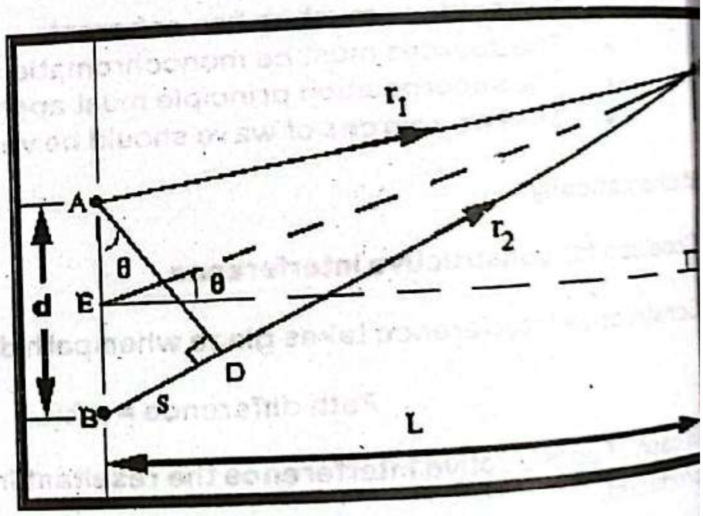
Chapter 13 Youngs Double Slit Experiment

YOUNG'S DOUBLE SLIT EXPERIMENT:

Introduction

The first practical demonstration of optical interference was provided by THOMAS YOUNG in 1801. His experiment gave a very strong support to the wave theory of light.

Experimental Arrrangement



'S' is a slit, which receives light from a source of monochromatic light. As 'S' is a narrow slit so it diffracts the light and it falls on slits A and B. After passing through the two slits, interference between two waves takes place on the screen. The slits A and B act as two coherent sources of light. Due to interference of waves alternate bright and dark fringes are obtained on the screen.

Quantitative Analysis

Let the wave length of light = λ

Distance between slits A and B = d

Distance between slits and screen = L

Consider a point 'P' on the screen where the light waves coming from slits A and B interfere such that PC=y. The wave coming from A covers a distance $AP=r_1$ and the wave coming from B covers distance $BP=r_2$ such that PB is greater than PA.

$$\text{Path difference} = BP - AP = BD$$

$$S = r_2 - r_1 = BD$$

In right angled $\triangle BAD$

$$\sin\theta = \frac{BD}{AB}$$

Or

$$\sin\theta = \frac{S}{d} \text{ .... (1)}$$

In right angled $\triangle PEC$

$$\sin\theta = \frac{PC}{PE}$$

$$\sin\theta = \frac{Y}{PE}$$

In this experimental setup PE $\cong$ CE, So PE = L

Now

$$\sin\theta = \frac{Y}{L} \text{ -------- (2)}$$

Compare equation (1) and equation (2) we get

$$\frac{Y}{L} = \frac{S}{d}$$

Or

$$Y = S\frac{L}{d}$$

For Bright Fringes OR Constructive Interference:

For bright fringe $S = m\lambda$

$$Y = m\lambda\frac{L}{d}$$

Where m = 0,1,2,3 ....

For Dark Fringes OR Destructive Interference

For dark fringes $S = (m + \frac{1}{2})\lambda$

$$Y = (m + \frac{1}{2})\lambda\frac{L}{d}$$

Where m = 0,1,2,3 ....

Fringe Spacing

The distance between any two consecutive bright fringes or two consecutive dark fringes is called fringe spacing.

Fringe spacing or thickness of a dark fringe or a bright fringe is equal. It is denoted by Δx

Consider a bright fringe

For bright fringe n = 1

$$Y_1 = \lambda\frac{L}{d}$$

For bright fringe n =2

$$Y_2 = 2\lambda \frac{L}{d}$$

Fringe spacing

$$\Delta x = Y_2 - Y_1$$

$$\Delta x = 2\lambda \frac{L}{d} - \lambda \frac{L}{d}$$

$$\Delta x = \lambda \frac{L}{d}$$

Similar result can be obtained for dark fringe.

Conclusion
Young's double slit experiment support the wave nature of light.

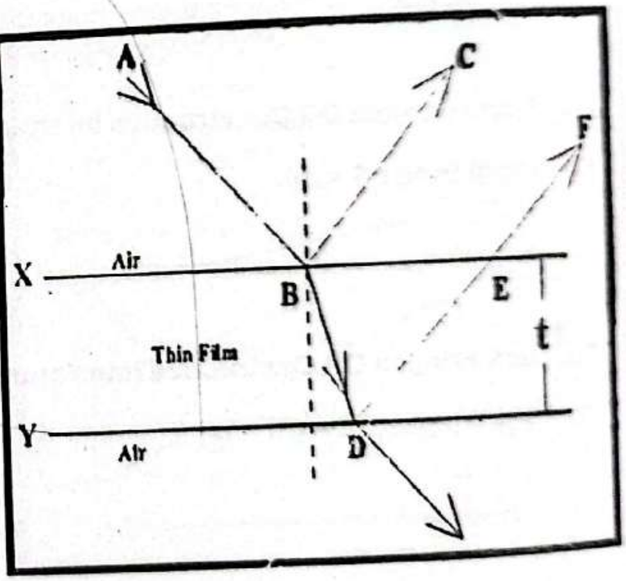
Chapter 13 | Interference in thin film

INTERFERENCE IN THIN FILM:

Thin Films
A film whose thickness is comparable with the thickness of light fallen is known as thin film.

Examples
Soap bubbles and thin layers of oil floating on water surface are common examples of thin films. When light is reflected from such a thin film, we observe different colors.

Interference in Thin Films
In thin films of soap and oil, visible color bands are the result of interference of light in thin films. The interference in this situation is caused by the interference of light waves reflected from the opposite surfaces of thin films. White light consists of seven colors each of different wave length; therefore, interference may be constructive or destructive.



ANALYSIS OF INTERFERENCE IN THIN FILMS

To understand the interference of light in thin films, consider two plates X & Y of ultra thin glass separated by a thin uniform film of air as shown in the diagram.
The film has a refractive index equal to 'n' and is surrounded by air whose refractive index is equal to one.

The ray of light ' AB ' strikes the upper surface of film at point B on the air-glass inter phase. The ray ' AB ' is partially reflected as a ray ' BC ' and partially transmitted as ray ' BD '. At point D another partial reflection takes place as ray ' EF '. The ray BC & EF interfere each other and produce

constructive and destructive interference depending upon the phase relation.
For nearly normal incidence,

$$\text{path difference} = 2t$$

If path difference is an integral multiple of λ, we expect constructive interference otherwise destructive interference. Unfortunately here the situation is different. First we consider what happened to the phase of rays which are reflected and transmitted. We also consider that two wave lengths are involved here.
The wave length in medium is

$$\lambda_n = \frac{\lambda}{n}$$

What happens to phase of rays?
When uniform thickness of air film is converted into a thin film of variable thickness along the line where plates are in contact, there is no path difference and we expect a constructive interference. But in actual practice this does not happen.
One of the two rays has gone under a phase change of 180° during reflection and therefore, conditions for constructive and destructive interference are reversed. Here we observe that only ray BC undergoes phase reversal. Ray BC and EF which are out of phase and interfering each other.
Path difference in case of thin film for constructive interference is:

$$\text{Path difference} = \left(m + \frac{1}{2}\right)\lambda_n$$

But

$$\text{Path difference} = 2t$$

Thus

$$2t = \left(m + \frac{1}{2}\right)\lambda_n$$

From equation (1) $\lambda_n = \lambda/n$

$$2t = \frac{\left(m + \frac{1}{2}\right)\lambda}{n}$$

$$2nt = \left(m + \frac{1}{2}\right)\lambda \qquad \text{Where } m = 0, 1, 2, 3, 4, \ldots$$

Similarly for destructive interference

$$2nt = m\lambda \qquad \text{Where } m = 0, 1, 2, 3, 4, \ldots$$

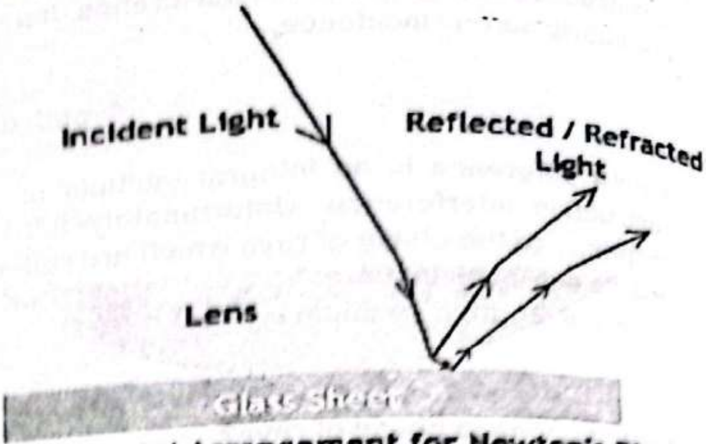
Position of bright and dark fringes depends on the wave length of light. When white light is used, colored fringes are obtained.

Chapter 13 — **Newtons Ring**

NEWTONS RING:
This phenomena was firstly observed by Sir Isaac Newton in 1717. It is based on the phenomena of interference.

Definition

When a Plano convex lens of large focal length is placed on a plane glass sheet and a parallel beam of light is made to fall on the plane surface of lens, on the sight from above with the help of microscope will give alternate dark and bright fringes in concentric circle. These circles are known as Newton's rings



Experimental Arrangement for Newton's Rings

FAILURE OF CORPUSCULAR THOERY:
Corpuscular theory was failed to explain this phenomena knowing this fact that corpuscular theory was also purposed by Newton.

Explanation

- An air film is trapped between the glass sheet and the convex surface of the lens, the thickness of air film is not constant it can be varied.
- The thickness of film increases in all directions as we move away from the point of contact of lens and glass sheet.
- When a light ray strikes the inner concave surface of lens it is partially reflected as well as partially refracted.
- When a refracted ray strikes the plane glass it undergoes a phase change of 180° or phase reversal on reelection.
- Interference occurs between the two waves which interface constructively if their path difference is $(m+\frac{1}{2})\lambda$.

And destructive interference if path difference is $m\lambda$

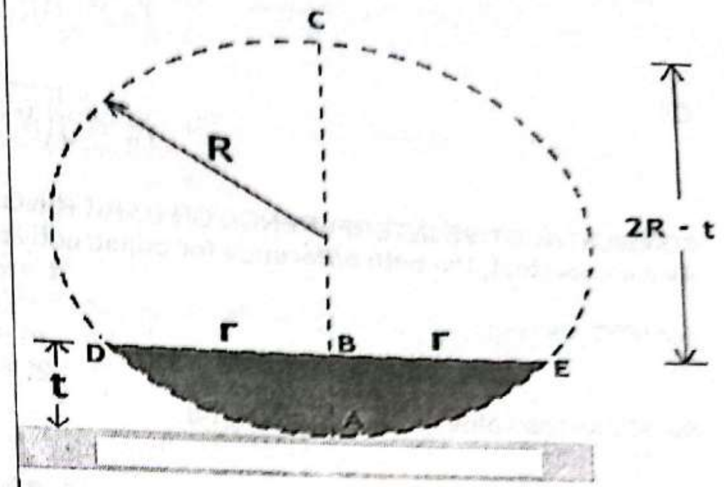
Where , m = number of fringes.

Theory

- Light ray I and II which reflected from the lower surface of lens and upper surface of glass plate superimpose on each other and superimpose.
- Due to their constructive and destructive interference dark and bright circular rings are formed.
- At the point of contact of lens and plate the path difference between the two rays is zero and as a change of 180° or phase reversal in the phase of light waves occur so they cancel out and hence a dark spot is formed at the center it means destructive interference.
- Path difference varies as one move away from central point, and thickness of air film increases.
- The fringes formed are circular because air film is symmetrical about the point of contact.

CALCULATION OF RADIUS OF RING:

Lets consider
R = Radius of curvature of lens
r = Radius of ring under consideration
t = Thickness of air film



Using the geometrical theorem that the product of intercepts of intersecting chords is equal, we have.

$$(DB) \times (BE) = (BC) \times (AB)$$

$DB = BE = r$,

$$r \times r = (BC) \times (AB)$$

$$r^2 = (BC) \times (AB) \text{ ——— (i)}$$

from diagram:

$$BC = 2R - t$$

& $AB = t$

equation (i) becomes

$$r^2 = (2R - t) \times t$$

$$r^2 = 2Rt - t^2$$

As R >>t and 't^2' being very small therefore it is neglected

$$r^2 = 2tR$$

$$r = \sqrt{2tR} \text{ ——— (ii)}$$

FOR CONSTRUCTIVE INTERFERENCE OR BRIGHT RINGS

As we know that, the path difference for constructive interference in this film is $(m + \frac{1}{2})\lambda$

Or

$$2t = (m + \frac{1}{2})\lambda \text{ ——— (iii)}$$

Where m = 0, 1, 2, 3.....

By substituting the value of "2t" from equation (iii) in equation (ii)

$$r = \sqrt{(m + \frac{1}{2})\lambda R} \text{ ——— (iv)}$$

For N^{th} bright ring

$$N = m + 1$$
$$m = N - 1$$

m	N
0	1
1	2
2	3
3	4
(N-1)	N

Substitute the value of m into equation (iv)

$$r_n = \sqrt{\left(N - 1 + \frac{1}{2}\right)\lambda R}$$

Or

$$r_n = \sqrt{\left(N - \frac{1}{2}\right)\lambda R} \quad \text{where } N = 1,2,3,4\ldots$$

FOR DESTRUCTIVE INTERFERENCE OR DARK RINGS

As we know that, the path difference for constructive interference in this film is

$$2t = m\lambda$$

For N^{th} Dark ring,

$$m = N$$
$$2t = N\lambda$$

Substitute the value of 2t in equation (ii)

$$r_n = \sqrt{RN\lambda} \quad \text{where } N = 1,2,3,4\ldots$$

Chapter 13 **Michelson Interferometer**

MICHELSON'S INTERFEROMETER:

Introduction

In 1881 the American physicist Albert Abraham Michelson constructed the interferometer used in the Michelson-Morley experiment.

Application

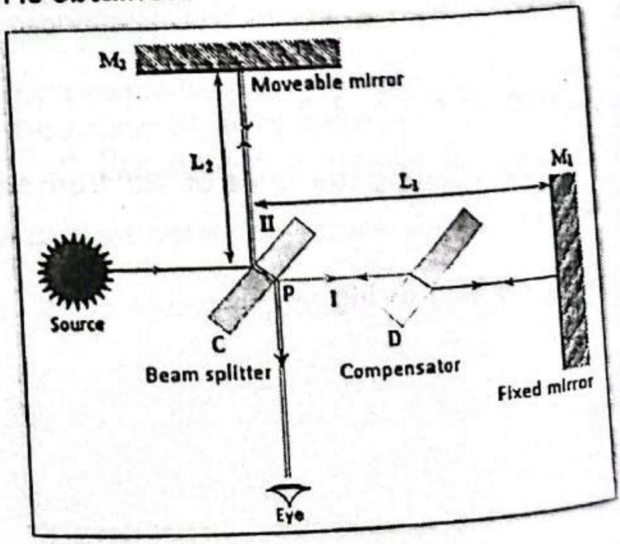
- This can be used to calculate the wave length of monochromatic light.
- This can be used to verify the wave nature of light.
- This can be used to observe interference of light.

Basic Principle

The light beam divided into two parts of equal intensity by partially reflection and refraction. These coherent light waves recombine and the interference pattern is obtained.

Construction

- It consists of a strong source of monochromatic light wave (λ).
- A semi polished mirror which is also used as a beam splitter (c). It is placed at 45°.
- Two full polished mirrors M_1 (Fixed) and M_2 (Moveable), initially placed equidistance from C initially.
- A glass plate identical to the beam splitter but not silvered. It is used to ensure that the beam I and II pass through the same thickness. That is why it is called compensator (D).



Working

Light waves are fall on semi polished mirror C and it is split in to two waves I & II. They are reflected back from mirrors M_1 and M_2 and finally enter the eye after being reflected and refracted from mirror C. The two waves I & II interface in the region between C and eye.

- Initially when both the mirrors (M_1 and M_2) are at equidistant from P ,the two rays I and II enter eye after covering back equal distance it means

  Path difference = 0

  Hence initially bright fringe will be seen [Condition for constructive interference]

- If mirror M_2 is displaced by an amount $\lambda/4$ then the ray II will cover a distance $(\frac{\lambda}{4}+\frac{\lambda}{4})=\frac{\lambda}{2}$ Greater than ray I.

  Hence **Path difference** $= \frac{\lambda}{2}$. [Condition for destructive interference]

  So they will interface destructively and a dark fringe will be seen.

- If mirror M_2 is further displaced by an amount equal to $\frac{\lambda}{4}$,its net displacement from initial position become $\frac{\lambda}{2}$,hence ray II will cover a distance $(\frac{\lambda}{2}+\frac{\lambda}{2}) = \lambda$ Greater than ray I

  Hence **Path difference** $= \lambda$

  So they will interface constructively again and a bright fringe will be seen.

Conclusion

Thus each time mirror M_2 is displaced by an amount of a bright fringe will be observed

If

Total displacement of M_2 = d

Total number of bright fringes = m

Then

$$d = m\frac{\lambda}{2}$$

Chapter 13 DIFFRACTION

DIFFRACTION

"The bending of light around an obstacle is called diffraction".

OR

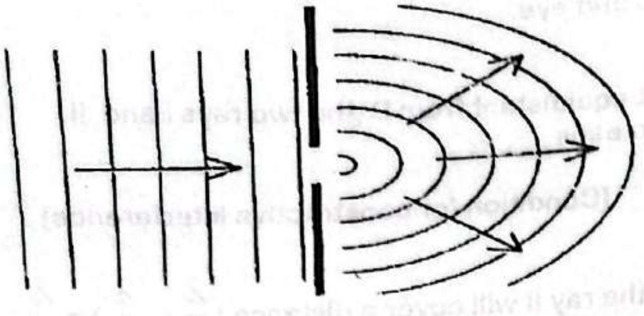
"The waves bend round the sides of an obstacle, or spread out as they pass through a gap. This effect is called Diffraction"

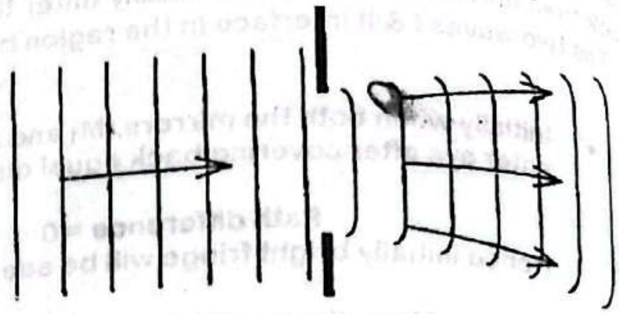
Diffraction is the special type of interference

Conditions For Diffraction

1. Source must have limited area.
2. Monochromatic light should be used.

3. Diffraction effect depends upon the size of object/obstacle. Diffraction takes place when size of obstacle is of the order of wavelength of the light used. Wider gaps produced less diffraction





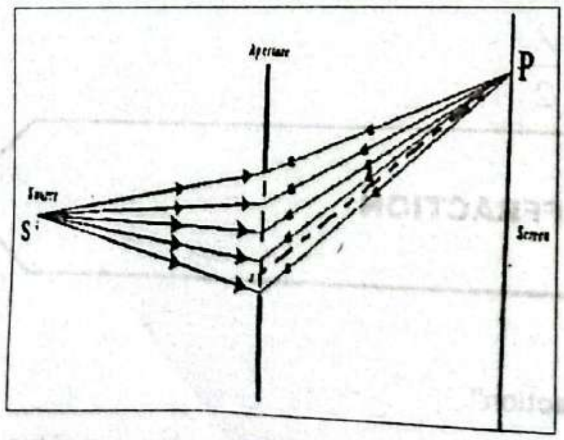
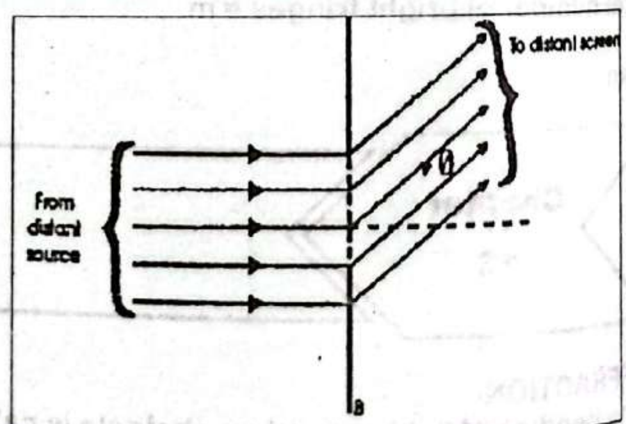
Diffraction Of Light Is Not Commonly Observed Phenomena

This is due to fact that light has very short wave lengths ranging (4×10^{-7} to 7×10^{-7}) m. If the size of obstacle is near to this limit of wave lengths then we observe diffraction.

Difference Between Interference And Diffraction Patterns

INTERFERENCE	DIFFRACTION
Interference is the result of interaction of light coming from two different wave fronts originating from the same source.	Diffraction is interaction of light coming from different parts of the same wave front.
The fringe spacing may or may not be of the same width.	Diffraction fringes are not of same width.
Points of minimum intensity are perfectly dark.	Points of minimum intensity are not perfectly dark.
All bright bands are of same intensity.	All bright bands are not of the same intensity.

Types Of Diffraction

FRESNEL DIFFRACTION	FRAUNHOFER DIFFRACTION
Source and screen are not far removed from each other.	Source and screen are far removed from each other..
	
Incident wave front is spherical.	Incident wave fronts are plane.
Wave front leaving the aperture or obstacle is also not parallel.	Diffraction obstacle gives rise to wave fronts which are also plane.
Lens is not used to make the rays parallel or to converge.	Plane diffracting wave fronts are converged by means of a convex lens to produce diffraction pattern.

DIFFRACTION GRATING

DIFFRACTION GRATING:

Definition

A diffraction grating is an optical device consists of a glass or polished metal surface over which thousands of fine, equidistant, closely spaced parallel lines are been ruled.

Principle

Its working principle is based on the phenomenon of diffraction. The space between lines act as slits and these slits diffract the light waves there by producing a large number of beams which interfere in such away to produce spectra.

Grating Element

Distance between two consecutive slits (lines) of a grating is called grating element. If 'a' is the separation between two slits and 'b' is the width of a slit, then grating element 'd' is given by:

$$d = a + b$$

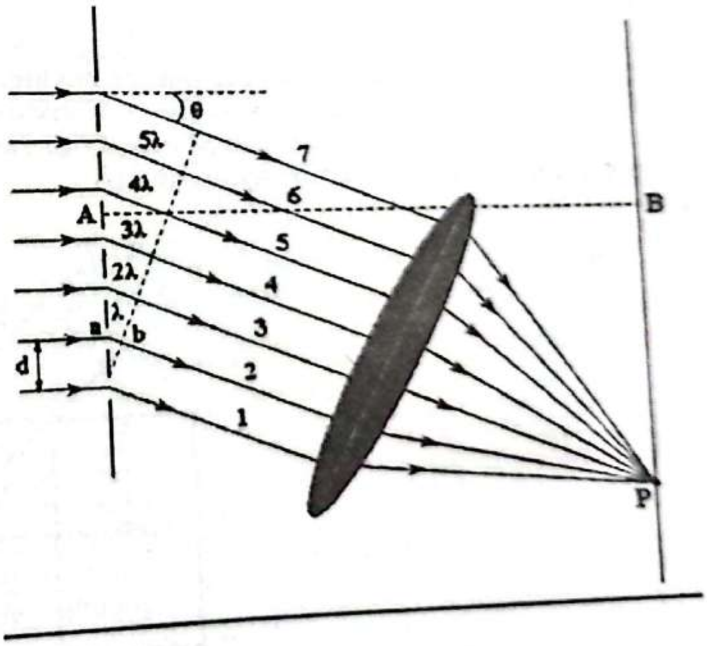
$$d = \text{length of grating / No. Of lines}$$

Or

$$d = \frac{L}{N}$$

DETERMINATION OF WAVE LENGTH BY DIFFRACTION GRATING

In diffraction grating, each ray travels a distance d sin q different from that of its neighbor, where d is the distance between slits. If this distance equals an integral number of wavelengths, the rays all arrive in phase, and constructive interference (a maximum) is obtained. Thus, the condition necessary to obtain constructive interference for a diffraction grating is:



$$m\lambda = d \sin\theta \qquad m=0,1,2,3,4.....$$

Where m = order of grating

Where d is the distance between slits in the grating, l is the wavelength of light, and m is the order of the maximum

This equation is called "grating equation" and is used to determine he wavelength of light.

Conclusion

- 'm' is called the order of grating and it is the number of bright or dark fringe obtained on the screen.

- For m=0, $\theta=0$, central bright maxima of zeroth order.
- For m=1, $\theta=\theta_1$, 1st order bright maxima (path difference $=\lambda$)
- For m=2, $\theta=\theta_2$, 2nd order bright maxima (path difference $= 2\lambda$)
- With the increase in 'm', fringes of decreasing width and less brightness are obtained.
- No order of line is possible at $\theta > 90^o$.

Characterstics Of Grating Spectra

- Spectra of different orders are obtained symmetrically on both sides of zeroth order image.
- Spectral lines are almost straight and quite sharp.
- Spectral colors are in the order.
- The spectral lines are more and more dispersed as we go to higher orders.
- Most of the incident intensity goes to zeroth order and rest of it is distributed among the other orders.

Chapter 13 — **X- Ray Diffraction**

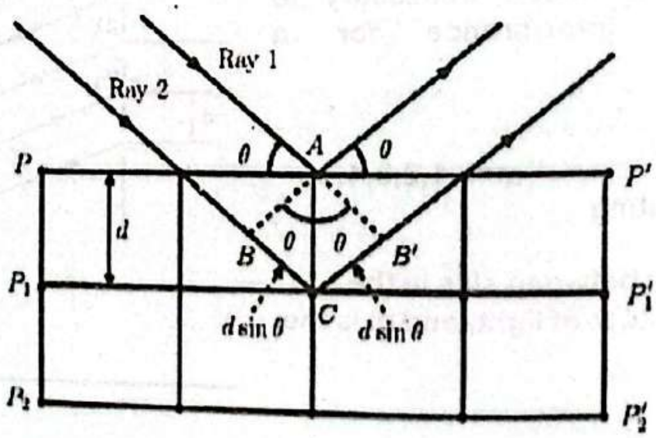
DIFFRACTION OF X-RAYS THROUGH CRYSTALS

The nature of x-rays is electromagnetic i.e. they are electromagnetic waves. X-rays have very short wavelength of the order of 10×10^{-10} m. Therefore it is not possible to produce interference fringes of x-rays by Young's double slit experiment or by thin film method. The reason is that the fringe spacing is $y = \lambda L/d$ and unless the slits are separated by a distance of 10×10^{-10} m, the fringes so obtained will be closed together that they can not be observed.

How ever it is possible to obtain x-rays diffraction by making use of crystals such as rock salt in which the atoms are uniformly spaced in planes and separated by a distance of order of 2 A° to 5 A°. Therefore, the diffraction of x-rays takes place when they incident on the surface of crystals.

BRAGG'S EQUATION

Consider a set of parallel lattice planes having spacing 'd' between each other as shown. Consider two rays 'Ray 1' and 'Ray 2' incident on the surface of crystal of NaCl. After reflection these rays reflected and are in phase. After reflection they interfere each other. The path difference between the two reflected rays is given by



Path difference = BC + B'C

As $\overline{BC} = \overline{B'C}$

Therefore,

Path difference $= \overline{BC} + \overline{BC}$

Path difference $= 2\overline{BC}$ .... (1)

In $\triangle ABC$

$$\frac{\overline{BC}}{\overline{AC}} = \sin\theta$$

$$\overline{BC} = \overline{AC}\sin\theta$$

$$\overline{BC} = d\sin\theta$$

Thus, in equation (1)
The path difference = $2\,d\sin\theta$

Now the X-rays will interfere constructively if the path difference is an integral multiple of wavelength
Thus,

$$m\lambda = 2\,d\sin\theta$$

Where "m" is the order of spectrum
This relation is known as Bragg's Law. The spacing of the atomic layers of crystals can be found from the density and atomic weight. Both 'm' and 'θ' can be measured and hence the wave length of rays can be measured by using Bragg's equation.

Limitation

Bragg's reflection will only be obtained when $\lambda \le 2d$ this is why we can't use visible light

Application Of Bragg's Law

Bragg's law is used in "x-ray crystallography". In this field the structure of solid is investigated by using x ray.

Chapter 13 Multiple Choice Questions

	MCQ'S	ANS.
1.	If the wavelength of an electromagnetic wave is about the diameter of a cricket ball, what type of radiation is it. a) X-ray b) Ultraviolet c) Radio waves d) Visible light	c
2.	Electromagnetic waves from an unknown source in space are found to be diffracted when passing through gaps of the order of 10^{-5} m, which type of the wave are they most likely to be? a) microwaves b) Ultraviolet c) Radio waves d) infra-red waves	a
3.	Huygens's conception of secondary waves a) Helps us to find the focal length of a thick lens b) is a geometrical method to find a wave front c) is used to determine the velocity of light d) is used to explain the polarization of light	b
4.	Interference fringes are produced using monochromatic light of same intensity from a double slit screen. If the intensity of light emerging from one of the slit is reduced, the effect on interference pattern will be a) All the dark and bright fringes become brighter. b) All the dark and bright fringes become darker. c) Bright fringes become brighter and dark fringes become darker. d) Bright fringes become darker and dark fringes become brighter.	d

5.	In Young's experiment when the distance between slits and screen is doubled, while separation of slits is halved, then fringe width will be; a) 4 times b) 1/4 times c) doubled d) unchanged	a
6.	A ray of light passes from air into water. Striking the surface of the water with an angle of incidence 45°. Which of these quantities change as the light enters the water.I) Wavelength ii) frequency iii) speed of propagation iv) direction of propagation a) i and ii only. b) i, iii, and iv only. c) iii and iv only, d) all of them.	b
7.	A hill separates a television (TV) transmitter from a house. The Transmitter cannot be seen from the house but still the TV in the house has good reception. What wave phenomena make it possible? a) Coherence of waves b) Interference of waves b) Diffraction of waves d) Refraction of waves	b
8.	Monochromatic light is incident on a diffraction grating and a diffraction pattern is observed. Which effects is observed by replacing the grating with one that has more lines per millimeter? a) Number of maxima decreases with decrease in angle between first and second order maxima. b) Number of maxima decreases with increase in angle between first and second order maxima. c) Number of maxima increases with decrease in angle between first and second order maxima. d) Number of maxima increases with increase in angle between first and second order maxima.	d
9.	Optically active substances are those substances which a) Produce polarized light b) rotate the plane of polarization of polarized light c) produce double refraction d) convert a plane polarized light into circularly polarized light	b
10.	Plane polarized light is passed through a Polaroid. On viewing through the Polaroid we find that when Polaroid is given one complete rotation about the direction of light a) The intensity of light gradually decreases to zero and remains at zero. b) The intensity of light gradually increases to maximum and remains at maximum. c) There is no change in the intensity of light. d) The intensity of light varies such that it is twice maximum and Twice zero.	d